Heart Disease Prediction Using Ensemble Machine Learning Methods

White Paper — Draft

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This white paper contains analysis performed on de-identified clinical data. All personally identifiable information was removed from the source dataset prior to use.

Business Problem

Cardiovascular disease (CVD) is the leading cause of death in the United States, responsible for approximately one in five deaths annually (CDC, 2024). Despite its prevalence, early-stage heart disease frequently goes undetected until a significant cardiac event occurs. Current diagnostic pathways rely heavily on invasive testing such as coronary angiography, stress testing, and imaging — procedures that are costly, time-intensive, and inaccessible to many patients, particularly in underserved communities.

This project addresses a practical clinical question: can a machine learning model trained on routine, non-invasive patient measurements reliably predict the presence of heart disease and thereby support earlier, lower-cost screening decisions? The business value is clear — a validated predictive tool could help clinicians triage higher-risk patients for accelerated diagnostic workup, reduce avoidable cardiac events, and allocate limited cardiology resources more efficiently.

The research questions guiding this analysis are: (1) Which clinical features are most predictive of heart disease presence? (2) Which machine learning algorithm produces the best balance of sensitivity and overall predictive performance? (3) How accurately can a model distinguish disease presence from absence using only intake-level measurements?

Background and History

The dataset used in this analysis originates from the Cleveland Clinic Foundation and was collected by Dr. Robert Detrano and colleagues in the late 1980s as part of an international collaborative effort to develop a probabilistic algorithm for coronary artery disease diagnosis (Detrano et al., 1989). The original database contains 76 attributes per patient; however, all published machine learning experiments have focused on a well-established subset of 14 features, which has become the de facto standard for benchmarking cardiac classification models. The dataset was subsequently donated to the UCI Machine Learning Repository, where it has been accessed and studied by researchers for over three decades.

Machine learning approaches to heart disease prediction have a substantial published history. Early work focused on decision trees and logistic regression given their interpretability in clinical contexts (Detrano et al., 1989). More recent research has explored ensemble methods, support vector machines, and deep learning, with reported AUC scores on the Cleveland dataset typically ranging from 0.85 to 0.93 depending on preprocessing choices, cross-validation strategy, and model selection. The present analysis contributes to this body of work by systematically comparing five classification algorithms using a rigorous cross-validation methodology and prioritizing recall — the clinical sensitivity to true disease cases — as the primary decision metric alongside AUC.

Data Explanation

Data Source and Preparation

The Cleveland Heart Disease dataset contains 303 patient records and 14 variables. Two fields — ca (number of major vessels colored by fluoroscopy) and thal (thalassemia result) — contained six missing values encoded as '?' in the raw file. These were replaced with each feature's median value, a standard approach for small proportions of missingness (6 of 303 records, approximately 2%). No other data quality issues were identified. The target variable, originally encoded as integers 0–4 representing disease severity, was binarized to 0 (no disease) and 1 (disease present, any severity) per the established experimental convention for this dataset. The resulting class distribution is well balanced: 164 (54.1%) negative and 139 (45.9%) positive, requiring no resampling.

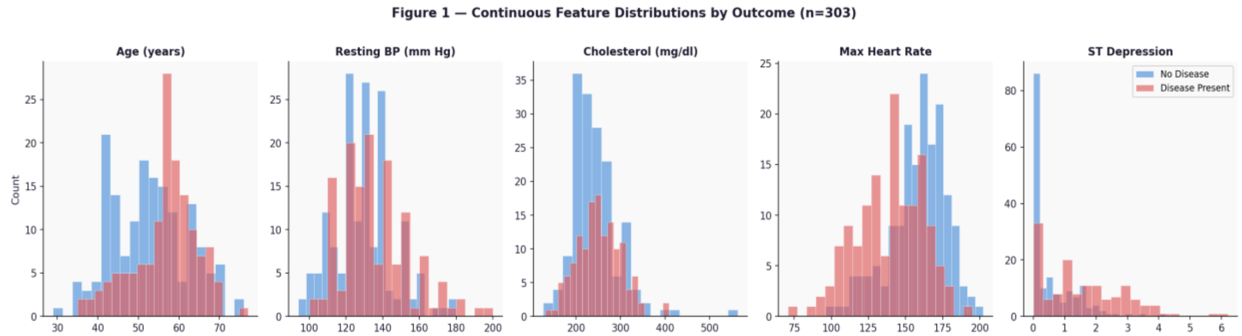

Figure 1. Distribution of continuous clinical features by heart disease outcome (n=303). Max heart rate (thalach) and ST depression (oldpeak) show the clearest class separation.

Data Dictionary

Feature	Type	Description
age	Continuous	Patient age in years (range: 29–77)
sex	Binary	Patient sex: 0 = Female, 1 = Male
cp	Categorical	Chest pain type: 1=Typical angina, 2=Atypical angina, 3=Non-anginal, 4=Asymptomatic
trestbps	Continuous	Resting blood pressure in mm Hg (range: 94–200)
chol	Continuous	Serum cholesterol in mg/dl (range: 126–564)
fbs	Binary	Fasting blood sugar > 120 mg/dl: 0 = No, 1 = Yes
restecg	Categorical	Resting ECG results: 0=Normal, 1=ST-T wave abnormality, 2=LV hypertrophy
thalach	Continuous	Maximum heart rate achieved (range: 71–202)
exang	Binary	Exercise-induced angina: 0 = No, 1 = Yes
oldpeak	Continuous	ST depression induced by exercise relative to rest (range: 0–6.2)
slope	Categorical	Slope of peak exercise ST segment: 1=Upsloping, 2=Flat, 3=Downsloping
ca	Ordinal	Number of major vessels colored by fluoroscopy (0–3; 4 values imputed)
thal	Categorical	Thalassemia result: 3=Normal, 6=Fixed defect, 7=Reversible defect (2 imputed)
target	Binary	Heart disease diagnosis (binarized): 0 = Absent, 1 = Present

Table 1. Data dictionary for the Cleveland Heart Disease dataset (14-feature subset).

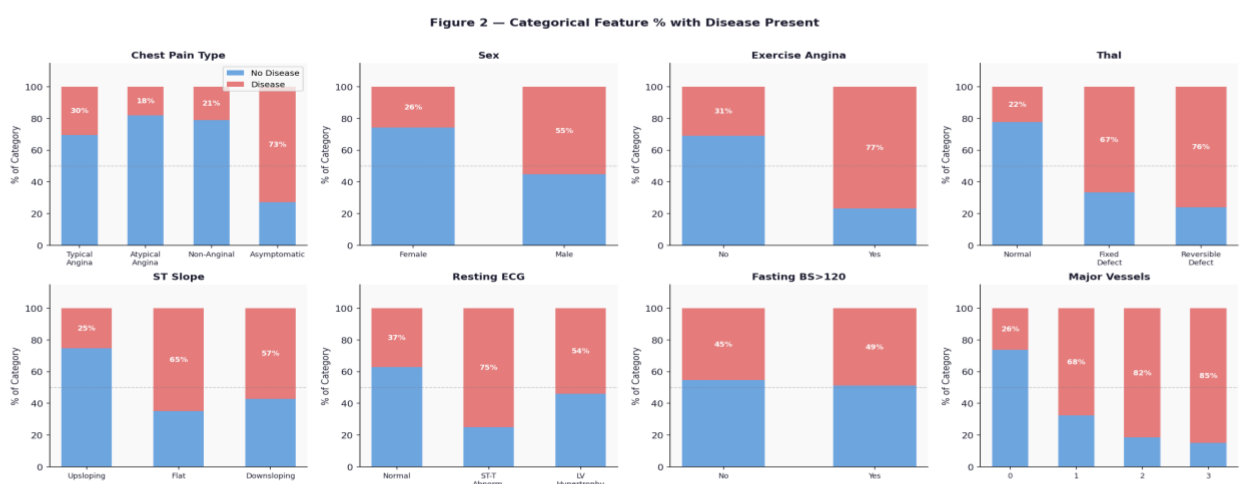

Figure 2. Proportion of heart disease present within each category of the eight categorical features. Asymptomatic chest pain (cp=4), exercise angina, and reversible thalassemia defect show the strongest associations.

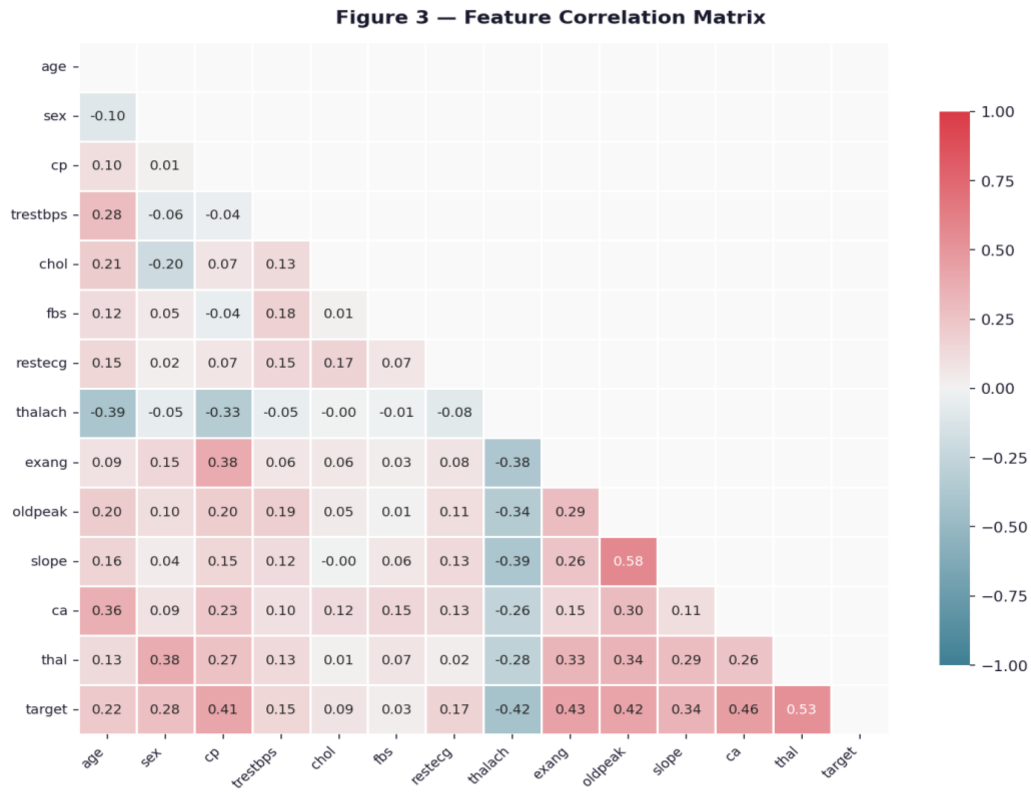

Figure 3. Pearson correlation matrix across all 14 features. thal, ca, exang, oldpeak, cp, and thalach show the strongest correlations with the target variable.

Methods

All analysis was conducted in Python 3.12 using pandas for data manipulation, scikit-learn for modeling, and matplotlib/seaborn for visualization. The analytical workflow proceeded across three phases.

Phase 1 — Exploratory Data Analysis

Initial EDA examined feature distributions, class balance, missing value patterns, and pairwise correlations. Histograms stratified by outcome class were produced for all continuous features. Stacked bar charts were used to visualize the proportion of disease-present patients within each category of the eight categorical features. A Pearson correlation heatmap was computed across all features to identify multicollinearity and the strongest individual predictors of the target.

Phase 2 — Model Development and Comparison

Five classification algorithms were evaluated: Logistic Regression, Random Forest, Gradient Boosting, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). Features requiring scale sensitivity (Logistic Regression, SVM, KNN) were wrapped in a scikit-learn Pipeline with StandardScaler normalization applied within each cross-validation fold to prevent data leakage. All models were evaluated using stratified 5-fold cross-validation. Performance metrics reported are cross-validated means for accuracy, recall (sensitivity), F1-score, and AUC-ROC. Recall was designated the primary clinical metric, as false negatives — predicting no disease in a patient who has it — carry greater clinical risk than false positives in a screening context (Siegel, 2016).

Phase 3 — Hyperparameter Tuning and Ensemble

The three highest-performing models (Logistic Regression, Random Forest, Gradient Boosting) were tuned using GridSearchCV with 5-fold cross-validation and AUC-ROC as the optimization metric. The tuned models were then combined into a soft-voting ensemble classifier, which predicts class probabilities as the average of individual model probabilities. Feature importances were extracted from the tuned Random Forest, and coefficient values were extracted from the tuned Logistic Regression to support clinical interpretation.

Analysis

Table 2 summarizes 5-fold cross-validated performance across all five models plus the tuned ensemble. Logistic Regression and Random Forest tied for the highest AUC at 0.912, with Logistic Regression achieving the highest recall at 0.791 — meaning it correctly identified approximately 79% of true disease cases. SVM achieved the highest raw accuracy (0.838) and F1-score (0.815), but its AUC of 0.898 trailed the top two models. Gradient Boosting underperformed relative to expectations, likely due to the small dataset size (n=303) limiting its ability to benefit from boosting's sequential error correction. The tuned ensemble (0.908 AUC) did not outperform the best individual models, a common outcome when dataset size limits ensemble diversity.

Model	Accuracy	Recall	F1 Score	AUC-ROC
Logistic Regression	0.832	0.791	0.813	0.912
Random Forest	0.822	0.770	0.799	0.912
SVM	0.838	0.784	0.815	0.898
Ensemble (Tuned)	0.835	0.776	0.811	0.908
KNN	0.822	0.777	0.800	0.882
Gradient Boosting	0.782	0.733	0.754	0.900

Table 2. Five-fold cross-validated performance metrics for all models. Recall prioritized as primary clinical metric.

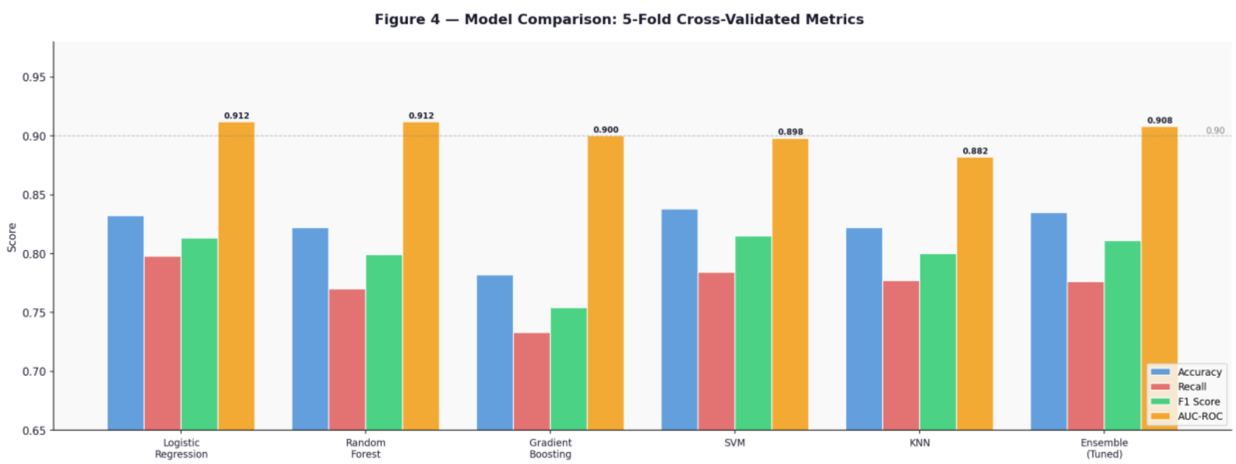

Figure 4. Cross-validated model comparison across four metrics. Logistic Regression and Random Forest achieve the highest AUC (0.912). Dashed line marks the 0.90 AUC threshold.

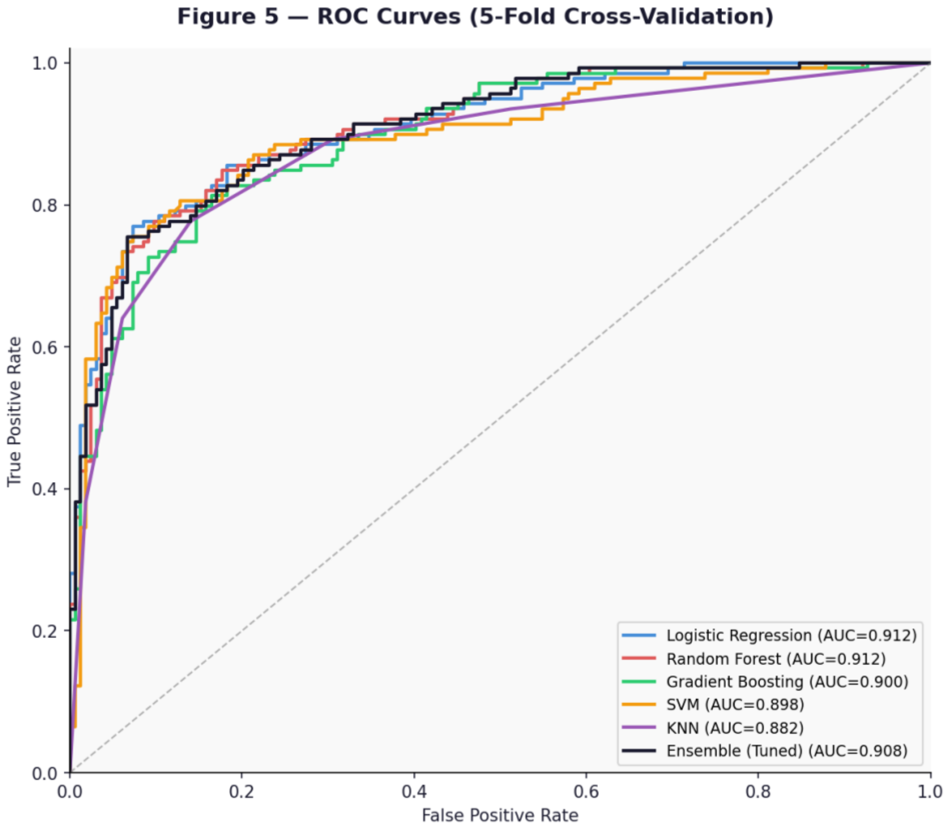

Figure 5. ROC curves for all six models under 5-fold cross-validation. Logistic Regression and Random Forest curves closely overlap near the top-left corner.

The confusion matrix analysis (Figure 6) further supports Logistic Regression as the preferred model. With a recall of 0.791 and a specificity of 0.878, it achieves the best balance between minimizing missed disease cases and avoiding unnecessary false alarms. Random Forest shows slightly lower recall (0.770) but higher specificity (0.902). SVM falls between the two on both metrics (recall=0.784, specificity=0.884).

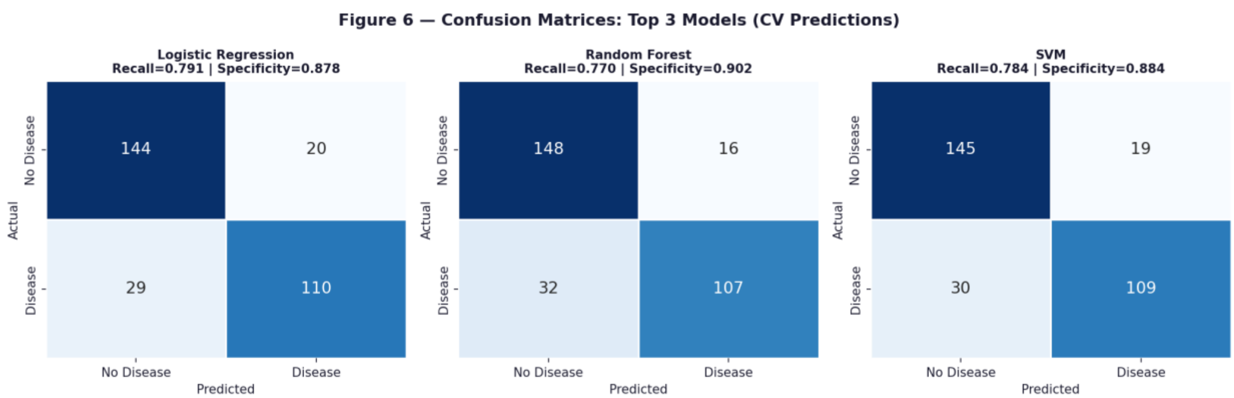

Figure 6. Confusion matrices for the top three models based on cross-validated predictions. Logistic Regression achieves the highest recall (0.791), minimizing missed disease cases.

Feature importance analysis (Figure 7) reveals consistent findings across both model types. Thalassemia result (thal), number of fluoroscopy-colored vessels (ca), chest pain type (cp), maximum heart rate (thalach), and ST depression (oldpeak) emerge as the five strongest predictors in both the Random Forest importance scores and the Logistic Regression coefficient magnitudes. Notably, fasting blood sugar (fbs) and serum cholesterol (chol) contribute negligibly to both models — a finding that aligns with prior literature suggesting these features have limited discriminative value in this dataset (Detrano et al., 1989).

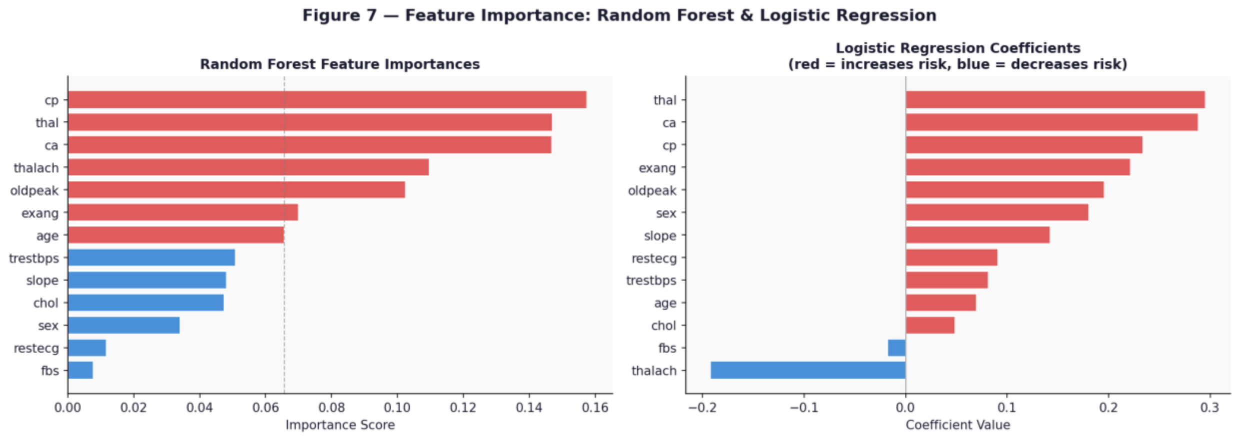

Figure 7. Feature importances from tuned Random Forest (left) and coefficient magnitudes from tuned Logistic Regression (right). thal, ca, cp, thalach, and oldpeak consistently rank as the top predictors across both model types.

Conclusion

This analysis demonstrates that machine learning models trained on routine clinical measurements can predict heart disease presence with strong accuracy and clinical utility. Logistic Regression is recommended as the primary model, achieving an AUC of 0.912 and a recall of 0.791 under 5-fold cross-validation — meaning it correctly identifies approximately 79% of true disease cases using only 13 non-invasive intake features. Random Forest serves as a complementary model, matching Logistic Regression on AUC while providing interpretable feature importance rankings that align with established clinical knowledge.

The five most predictive features — thal, ca, cp, thalach, and oldpeak — are consistent across both model types and with prior published work on this dataset, lending additional confidence to the findings. The near-zero predictive contribution of fbs and chol suggests these features may be safely deprioritized in future model iterations without meaningful loss of performance.

Assumptions

- The six missing values in ca and thal are missing at random and median imputation does not introduce meaningful bias into model training.
- Binarizing the target variable (collapsing severity values 1–4 into a single ‘disease present’ class) is an appropriate formulation for a screening use case where the clinical goal is detecting any degree of disease presence.
- The 14-feature subset represents a sufficiently comprehensive set of risk indicators for the population studied. The 62 excluded original attributes are assumed to contain either redundant or operationally unavailable information.
- Cross-validated performance metrics on this dataset are representative of how the model would perform on new patients drawn from a similar clinical population.

Limitations

- The dataset contains only 303 unique records, which limits the statistical power of model comparisons and may contribute to variance in performance estimates across cross-validation folds.

- The Cleveland cohort was collected in the 1980s and may not reflect current patient demographics, treatment standards, or measurement techniques. Model performance on a contemporary patient population has not been validated.
- The dataset is predominantly male (68%) and the demographic composition is not fully documented. Model performance may vary across sex and age subgroups in ways not detectable at this sample size.
- Collapsing severity levels 1–4 into a single binary label discards clinically meaningful information about disease severity that could be valuable in a full diagnostic context.
- Cholesterol values up to 564 mg/dl and a single patient with ca=4 suggest possible data entry anomalies that were not removed from the training data.

Challenges

- Threshold selection for the final model requires domain-informed judgment. The default 0.5 probability threshold may not be optimal; a lower threshold would increase recall at the cost of more false positives, a tradeoff that should be evaluated with clinical stakeholders.
- Gradient Boosting underperformed relative to its typical benchmark results, likely due to dataset size. Boosting methods generally require larger datasets to realize their sequential error-correction advantage.
- The '?' encoding for missing values was not documented in the UCI repository metadata and required manual inspection to detect. Future versions of this analysis should include automated missing-value detection across all encoding types.

Future Uses and Additional Applications

The modeling framework developed here is directly extensible to other cardiovascular risk prediction tasks. With a larger, more recent dataset — such as the MIMIC-IV clinical database or a de-identified EHR extract — the same pipeline could be retrained to produce a contemporary, validated screening tool. Multi-class severity prediction (distinguishing levels 0–4) represents a natural next step, enabling clinical stratification beyond binary presence/absence. Additionally, the feature importance findings could inform the design of a streamlined intake questionnaire that prioritizes the five highest-signal variables, reducing data collection burden in resource-constrained clinical settings.

Recommendations

- Deploy Logistic Regression as the primary screening model, citing its best-in-class recall (0.791), tied AUC (0.912), and full coefficient interpretability for clinical communication.
- Use Random Forest as a secondary validation model, leveraging its feature importance output to explain predictions to clinical staff.
- Evaluate classification threshold on a held-out validation set in partnership with clinicians to determine the operationally appropriate recall/specificity tradeoff before any real-world deployment.
- Retrain the model on a contemporary, larger dataset before any clinical use, and conduct prospective validation in a real patient cohort.
- Consider dropping fbs and chol in future model iterations given their negligible predictive contribution, simplifying the feature set without meaningful performance loss.

Implementation Plan

A phased implementation approach is recommended for any real-world deployment:

- Phase 1 (Months 1–3): Validate model on a contemporary de-identified EHR dataset from a target institution. Assess performance across demographic subgroups. Engage clinical stakeholders to establish an acceptable recall threshold.
- Phase 2 (Months 4–6): Develop a lightweight clinical decision support interface that accepts the 13 feature inputs and returns a risk score and probability estimate. Integrate with existing EHR workflow for pilot use.
- Phase 3 (Months 7–12): Conduct a prospective pilot study in a single outpatient cardiology or primary care clinic. Track model-flagged patients through diagnostic workup. Collect outcomes data for ongoing model refinement.
- Ongoing: Establish a model monitoring process to detect performance drift as patient population characteristics or clinical measurement standards evolve.

Ethical Assessment

Several ethical considerations must be addressed before this model is applied in a clinical setting:

- De-identification and privacy. Although patient names and social security numbers have been removed from the dataset, the combination of age, sex, and detailed clinical measurements may retain quasi-identifying characteristics. The dataset must not be joined to any external data source that could re-identify individuals.
- Demographic bias and equity. The Cleveland cohort is approximately 68% male and was collected at a single institution in the 1980s. Heart disease presentation and risk factor profiles differ by sex, age, race, and socioeconomic status. A model trained exclusively on this cohort may underperform for underrepresented groups. Fairness audits across demographic subgroups are required before clinical deployment.
- False negative consequences. In a cardiac screening context, a false negative — predicting no disease in a patient who has it — carries potentially life-threatening consequences. Model threshold selection must prioritize minimizing false negatives, and clinicians must understand that a negative model prediction is not equivalent to a clinical ruling-out of heart disease.
- Clinical decision support, not replacement. This model is designed to augment clinical judgment, not replace it. No patient management decision should be based solely on model output. Results should be communicated to clinicians as probabilistic risk signals requiring further evaluation.
- Transparency and explainability. Logistic Regression’s coefficient-based interpretability supports clinician trust and regulatory scrutiny. Any move toward more complex models in future iterations must be accompanied by appropriate explainability tooling (e.g., SHAP values) to maintain transparency.

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Appendix A — Supporting Documentation

A.1 — Model Hyperparameter Tuning Results

Model	Parameter	Search Space	Best Value
Random Forest	n_estimators	200, 300, 400	300
Random Forest	max_depth	None, 5, 10	5
Random Forest	min_samples_split	2, 5	5
Gradient Boosting	n_estimators	100, 200, 300	100
Gradient Boosting	learning_rate	0.05, 0.10, 0.20	0.05
Gradient Boosting	max_depth	2, 3, 4	2
Logistic Regression	C	0.01, 0.1, 1, 10, 100	0.01
Logistic Regression	penalty	l1, l2	l2

Table A1. GridSearchCV hyperparameter tuning results. Optimization metric: AUC-ROC, 5-fold CV.

A.2 — Software and Environment

Python 3.12 · pandas 2.x · scikit-learn 1.x · matplotlib 3.x · seaborn 0.13.x

A.3 — Cross-Validation Configuration

StratifiedKFold: n_splits=5, shuffle=True, random_state=42. Stratification on target variable ensures proportional class representation in each fold. StandardScaler applied within pipeline to prevent data leakage for scale-sensitive models.

A.4 — Missing Value Summary

ca: 4 records encoded as '?' → imputed with median (median=0.0). thal: 2 records encoded as '?' → imputed with median (median=3.0). All imputation performed prior to train/test splitting within the CV pipeline.